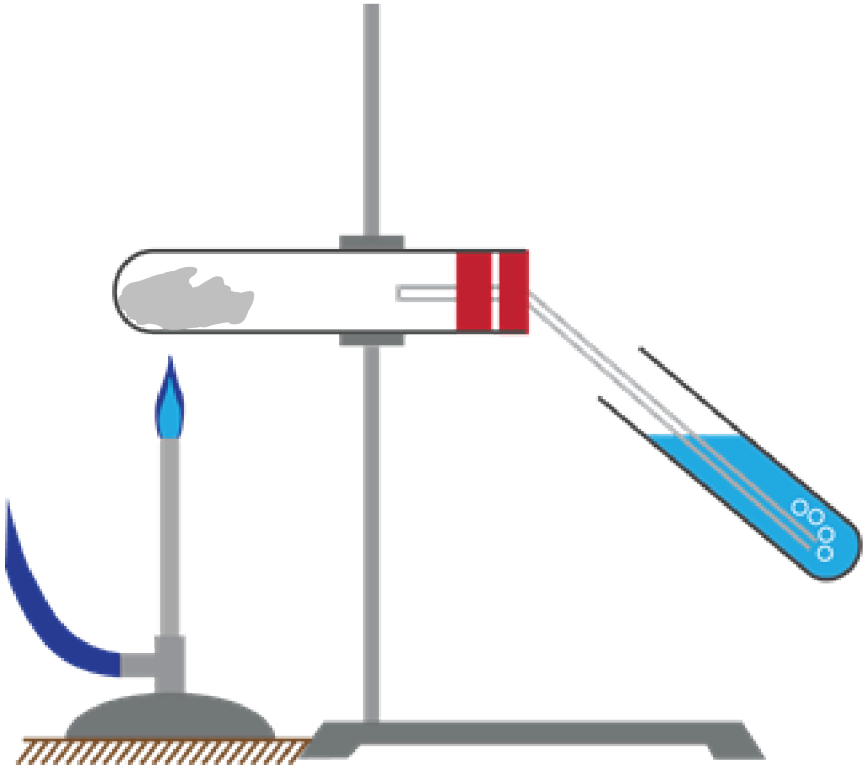
Sometimes when chemicals are mixed they react to form a new substance and sometimes a chemical will just need heat to break apart.

The word **thermal** means heat and another way of saying breaking down is '**decompose**' – so **thermal decomposition** means to break down with heat.

One example of this happening is chalk. The chemical name for chalk is calcium carbonate and when heated it breaks down into a chemical called calcium oxide and a gas called carbon dioxide.



This is the equipment that your teacher will have used to show you thermal decomposition.

Label the diagram using these words:

Bunsen burner	heat-proof mat	limewater
bung	delivery tube	stand and clamp

List three things that could injure or hurt someone during the demonstration.

.....

.....

.....

What observations did you make during the demonstration?

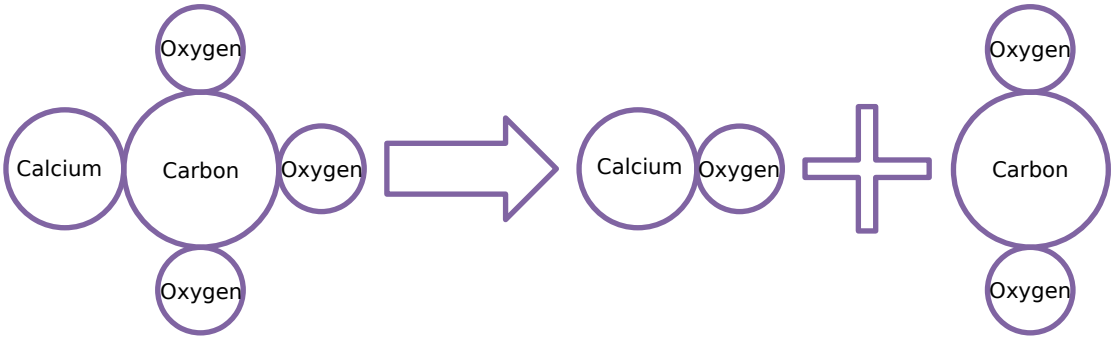
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Calcium carbonate breaks down (decomposes) with heat, it **thermally decomposes.**



1. Colour the calcium yellow, the carbon black, and the oxygen blue.

2. Before the reaction there is/are ..... compound(s).

   After the reaction there is/are ..... compound (s).

3. What has caused the compound(s) to break up? ..... .

4. What is this process called? T..... d.....

5. How many carbon atoms are there before thermal decomposition? .....

   How many carbon atoms are there after thermal decomposition? .....

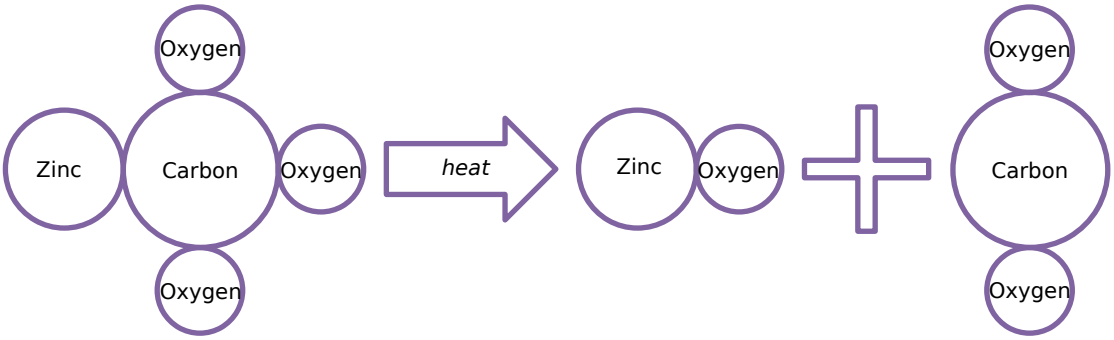
   How many oxygen atoms are there before thermal decomposition? .....

   How many oxygen atoms are there after thermal decomposition? .....

   How many metal atoms are there before thermal decomposition? .....

   How many metal atoms are there after thermal decomposition? .....

Zinc carbonate breaks down (decomposes) with heat, it **thermally decomposes.**



6. Colour the calcium yellow, the carbon black, and the oxygen blue.

7. Before the reaction there is/are ..... compound(s).

   After the reaction there is/are ..... compound (s).

8. What has caused the compound(s) to break up? ..... .

9. What is this process called? T..... d.....

10. How many carbon atoms are there before thermal decomposition? .....

    How many carbon atoms are there after thermal decomposition? .....

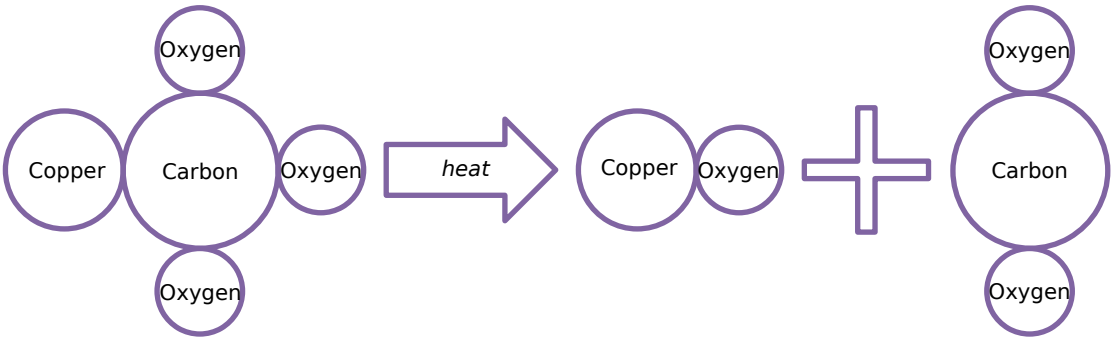
    How many oxygen atoms are there before thermal decomposition? .....

    How many oxygen atoms are there after thermal decomposition? .....

    How many metal atoms are there before thermal decomposition? .....

    How many metal atoms are there after thermal decomposition? .....

Copper carbonate breaks down (decomposes) with heat, it **thermally decomposes.**



11. Colour the copper green, the carbon black, and the oxygen blue.

12. Before the reaction there is/are ..... compound(s).

    After the reaction there is/are ..... compound (s).

13. What has caused the compound(s) to break up? ..... .

14. What is this process called? T..... d..... .

15. How many carbon atoms are there before thermal decomposition? ..... .

    How many carbon atoms are there after thermal decomposition? ..... .

    How many oxygen atoms are there before thermal decomposition? ..... .
    How many oxygen atoms are there after thermal decomposition? ..... .

    How many metal atoms are there before thermal decomposition? ..... .
    How many metal atoms are there after thermal decomposition? ..... .

Teaching notes and answers

This resource gives a good basis to explore the idea of conservation of atoms during a chemical reaction. Ask the students what they notice about the answers that they give to questions 5, 10 and 15.

The answers repeat themselves for questions 6-15.

2. Before the reaction there is one compound.

After the reaction there are two compounds.

3. Heat energy

4. Thermal decomposition

5. How many carbon atoms are there before thermal decomposition?	**One**
How many carbon atoms are there after thermal decomposition?	**One**
How many oxygen atoms are there before thermal decomposition?	**Three**
How many oxygen atoms are there after thermal decomposition?	**Three**
How many metal atoms are there before thermal decomposition?	**One**
How many metal atoms are there after thermal decomposition?	**One**